Technical Requirements Document

1. The system shall connect to Azure DevOps using REST APIs.

Technical Design (TRD)

- Integrate Azure DevOps REST APIs for data retrieval.
- Implement API connection validation and error handling.

Development Tasks

1. Develop Azure DevOps API integration service.
2. Configure connection validation and logging.

2. The system shall synchronize work items, user stories, tasks, bugs, and epics.

Technical Design (TRD)

- Build a synchronization service to fetch and process work items.

Development Tasks

1. Develop work item synchronization module.
2. Implement data mapping and duplicate validation.

3. The system shall store synchronized data in Azure SQL Database.

Technical Design (TRD)

- Design Azure SQL database schema for project data.
- Implement data persistence and retrieval services.

Development Tasks

1. Create database tables and relationships.
2. Develop data storage and retrieval logic.

4. The system shall authenticate users using Microsoft Entra ID.

Technical Design (TRD)

- Integrate Microsoft Entra ID using OAuth 2.0
- Implement secure user session management.

Development Tasks

1. Configure Entra ID authentication.
2. Implement login and logout functionality.

5. The system shall provide role-based authorization.

Technical Design (TRD)

- Implement Role-Based Access Control (RBAC).
- Define permissions for each user role.

Development Tasks

1. Create roles and permission mappings.
2. Implement authorization rules across modules.

6. The system shall prevent unauthorized access to protected REST API endpoints.

Technical Design (TRD)

- Secure APIs using token / Id based authentication.
- Enforce authorization checks before API access.

Development Tasks

1. Implement API authentication and authorization middleware.
2. Configure security validation for protected endpoints.

7. The system shall calculate sprint velocity, measure sprint completion rate, track defect density, calculate backlog health metrics, measure story completion trends, calculate release success rates, and provide KPI trend analysis.

Technical Design (TRD)

- Develop a KPI Engine to process synchronized Azure DevOps data and calculate delivery metrics.
- Store KPI results and historical trends for reporting and dashboard visualization.

Development Tasks

1. Implement KPI calculation services and business rules.
2. Develop KPI trend analysis and historical data storage.

8. The system shall display sprint progress, track completed versus planned work, identify delayed work items, and monitor sprint commitments.

Technical Design (TRD)

- Build a Sprint Governance module to monitor sprint execution and delivery performance.
- Generate sprint health indicators using sprint and work item data.

Development Tasks

1. Develop sprint tracking and progress monitoring services.
2. Implement delayed work item detection and sprint health calculations.

9. The system shall analyze delivery metrics using Azure OpenAI, identify delivery risks, predict schedule slippages, identify productivity bottlenecks, and recommend mitigation actions.

Technical Design (TRD)

- Integrate Azure OpenAI to evaluate delivery metrics and generate risk insights.
- Use AI prompts and governance data to produce risk predictions and recommendations.

Development Tasks

1. Implement Azure OpenAI integration and prompt management.
2. Develop risk analysis, prediction, and recommendation services.

10. The system shall generate executive summaries, summarize project health, generate portfolio reports, create sprint review summaries, and provide AI-generated recommendations.

Technical Design (TRD)

- Build an AI reporting module to transform governance data into business-friendly summaries.
- Generate downloadable reports and executive-level insights.

Development Tasks

1. Develop AI-powered report generation services.
2. Implement report export and summary generation functionality

11. The system shall provide role-based dashboards, display KPI metrics visually, display risk indicators, provide project health dashboards, and support filtering and drill-down capabilities.

Technical Design (TRD)

- Develop interactive dashboards with role-based data access and visual analytics.
- Implement filtering, drill-down, and data exploration capabilities.

Development Tasks

1. Create dashboard UI components and visualization widgets.
2. Implement filtering, drill-down, and role-based dashboard access.